Fixed-Anchor Tikhonov Continuation for Locally Lipschitz Monotone Inclusions: A Rigorous Partial-Progress Note

Prepared in response to a request to verify and salvage a proposed proof

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Status. This note does *not* prove a solution of Lu–Mei's open problem. It proves a clean fixed-anchor residual-transfer lemma and an $O(\varepsilon^{-1})$ continuation theorem conditional on a residual-warm-start version of the strongly monotone solver. The conditional hypothesis is precisely the missing step in the proposed proof being examined. The published Lu–Mei theorem gives a distance-based strong-solver bound; substituting that theorem into the fixed-anchor scheme recovers only the known $O(\varepsilon^{-1} \log \varepsilon^{-1})$ bound.

Abstract

Lu and Mei ask whether their $O(\varepsilon^{-1} \log \varepsilon^{-1})$ operation complexity for non-strongly monotone inclusions with a locally Lipschitz single-valued part can be improved to $O(\varepsilon^{-1})$. We examine a natural fixed-anchor Tikhonov continuation idea. For a fixed anchor a, the regularized inclusions

$$0 \in F(x) + B(x) + \mu(x - a), \qquad \mu \downarrow 0,$$

are strongly monotone. We prove that exact and approximate points on this path remain uniformly bounded and that a residual certificate for the μ_{s-1}-subproblem automatically gives a residual certificate of order μ_s for the next μ_s-subproblem when μ_s/μ_{s-1} is fixed. Hence, if one had a strongly monotone routine whose work depends on the logarithm of the *initial residual* divided by the target residual, the logarithmic factor would be constant at each continuation stage and the total work would be $O(\varepsilon^{-1})$. However, Lu–Mei's stated strong-solver complexity depends on an initial distance and the target tolerance. With that theorem, the fixed-anchor argument still gives $O(\varepsilon^{-1} \log \varepsilon^{-1})$, not $O(\varepsilon^{-1})$. The result is therefore partial progress and a precise reduction, not a resolution of the open problem.

1 Problem setting

We work in $\mathbb{R}^n$ with the Euclidean norm. Let

$$B : \mathbb{R}^n \rightrightarrows \mathbb{R}^n$$

be maximal monotone with nonempty domain, and let

$$F : \overline{\operatorname{dom} B} \to \mathbb{R}^n$$

be single-valued, monotone, and locally Lipschitz on $\overline{\operatorname{dom} B}$. Set

$$T := F + B.$$

As in Lu–Mei's setting, we assume that T is maximal monotone and that

$$\mathcal{S} := \operatorname{zer} T = \{x \in \operatorname{dom} B : 0 \in T(x)\}$$

is nonempty. For $x \in \operatorname{dom} B$, define the residual

$$\operatorname{res}_T(x) := \operatorname{dist}(0, T(x)) = \inf\{\|v\| : v \in T(x)\}.$$

An ε-residual solution is a point $x \in \operatorname{dom} B$ satisfying $\operatorname{res}_T(x) \le \varepsilon$. The operation model counts evaluations of F and resolvent calls

$$J_{\gamma B} := (I + \gamma B)^{-1}, \qquad \gamma > 0.$$

The resolvent is assumed exactly computable, matching Lu–Mei's model.

Lu–Mei prove an $O(\log \varepsilon^{-1})$ operation bound in the strongly monotone case and an $O(\varepsilon^{-1} \log \varepsilon^{-1})$ operation bound in the merely monotone case, and they ask whether the latter can be improved to $O(\varepsilon^{-1})$ [1]. The goal here is not to claim such a solution, but to isolate exactly what the fixed-anchor continuation idea proves and exactly where the missing ingredient lies.

2 Fixed-anchor regularization

Fix an anchor

$$a \in \operatorname{dom} B.$$

For $\mu > 0$, define the anchored Tikhonov operator

$$T_\mu(x) := T(x) + \mu(x - a) = F(x) + B(x) + \mu(x - a).$$

Since T is maximal monotone, the zero of T_μ exists and is unique: indeed,

$$0 \in T(p_\mu) + \mu(p_\mu - a) \quad \Longleftrightarrow \quad a \in p_\mu + \mu^{-1} T(p_\mu),$$

so

$$p_\mu = J_{\mu^{-1}T}(a).$$

Fix once and for all a solution $x^\star \in \mathcal{S}$, and write

$$D := \|a - x^\star\|.$$

The next two lemmas are the main unconditional facts.

Lemma 2.1 (Uniform bound for exact anchored solutions)**.** *For every $\mu > 0$, the unique zero p_μ of T_μ satisfies*

$$\|p_\mu - x^\star\| \le D, \qquad \|p_\mu - a\| \le 2D.$$

Proof. Choose

$$u_\mu \in T(p_\mu), \qquad u^\star = 0 \in T(x^\star).$$

The definition of p_μ gives

$$u_\mu = -\mu(p_\mu - a).$$

By monotonicity of T,

$$0 \le \langle u_\mu - u^\star, p_\mu - x^\star \rangle = -\mu \langle p_\mu - a, p_\mu - x^\star \rangle.$$

Thus

$$\langle p_\mu - a, p_\mu - x^\star \rangle \le 0.$$

Since

$$p_\mu - a = (p_\mu - x^\star) + (x^\star - a),$$

we get

$$\|p_\mu - x^\star\|^2 \le \langle a - x^\star, p_\mu - x^\star \rangle \le D \|p_\mu - x^\star\|.$$

If $p_\mu \ne x^\star$, division by $\|p_\mu - x^\star\|$ gives $\|p_\mu - x^\star\| \le D$; otherwise the inequality is trivial. The second estimate follows from the triangle inequality:

$$\|p_\mu - a\| \le \|p_\mu - x^\star\| + \|x^\star - a\| \le 2D.$$

□

Lemma 2.2 (Uniform bound and original residual for approximate anchored solutions). *Let $\sigma > 0$. Suppose $x \in \operatorname{dom} B$ and*

$$e \in T_\mu(x), \qquad \|e\| \le \sigma\mu.$$

Then

$$\|x - p_\mu\| \le \sigma, \qquad \|x - a\| \le 2D + \sigma.$$

Moreover

$$e - \mu(x - a) \in T(x)$$

and hence

$$\operatorname{res}_T(x) \le 2(D + \sigma)\mu.$$

Proof. Since $0 \in T_\mu(p_\mu)$ and T_μ is μ-strongly monotone,

$$\mu \|x - p_\mu\|^2 \le \langle e - 0, x - p_\mu \rangle \le \|e\| \; \|x - p_\mu\|.$$

This implies $\|x - p_\mu\| \le \|e\| / \mu \le \sigma$. Lemma 2.1 then gives

$$\|x - a\| \le \|x - p_\mu\| + \|p_\mu - a\| \le \sigma + 2D.$$

Finally, from $e \in T(x) + \mu(x - a)$, we have $e - \mu(x - a) \in T(x)$. Therefore

$$\operatorname{res}_T(x) \le \|e - \mu(x - a)\| \le \|e\| + \mu \|x - a\| \le \sigma\mu + \mu(2D + \sigma) = 2(D + \sigma)\mu.$$

□

3 The fixed-anchor residual transfer

Choose a geometric schedule

$$\mu_s := \mu_0 \theta^s, \qquad 0 < \theta < 1, \qquad s = 0, 1, 2, \ldots,$$

and solve the s-th anchored problem only to relative residual accuracy

$$\tau_s := \sigma \mu_s.$$

The crucial observation is that an approximate solution of stage $s - 1$ is already a residual warm start for stage s.

Lemma 3.1 (Stage-to-stage residual transfer)**.** *Assume that for stage $s-1$ we have*

$$e_{s-1} \in T_{\mu_{s-1}}(x^{s-1}), \qquad \|e_{s-1}\| \le \sigma\mu_{s-1}.$$

Then

$$e_s^{\mathrm{in}} := e_{s-1} - (\mu_{s-1} - \mu_s)(x^{s-1} - a)$$

satisfies

$$e_s^{\mathrm{in}} \in T_{\mu_s}(x^{s-1})$$

and

$$\left\|e_s^{\mathrm{in}}\right\| \le C_{\mathrm{w}}\mu_s,$$

where

$$C_{\mathrm{w}} := \frac{\sigma + (1-\theta)(2D+\sigma)}{\theta}.$$

Consequently

$$\frac{\left\|e_s^{\mathrm{in}}\right\|}{\tau_s} \le \frac{C_{\mathrm{w}}}{\sigma},$$

a constant independent of s and independent of the final tolerance ε.

Proof. Write

$$e_{s-1} = u_{s-1} + \mu_{s-1}(x^{s-1} - a)$$

for some $u_{s-1} \in T(x^{s-1})$. Then

$$e_s^{\mathrm{in}} = u_{s-1} + \mu_s(x^{s-1} - a) \in T_{\mu_s}(x^{s-1}).$$

By Lemma 2.2, applied to x^{s-1} and μ_{s-1},

$$\left\|x^{s-1} - a\right\| \le 2D + \sigma.$$

Since $\mu_s = \theta\mu_{s-1}$,

$$\begin{aligned}
\left\|e_s^{\mathrm{in}}\right\| &\le \|e_{s-1}\| + (\mu_{s-1} - \mu_s)\left\|x^{s-1} - a\right\| \\
&\le \sigma\mu_{s-1} + (1-\theta)\mu_{s-1}(2D+\sigma) \\
&= \frac{\sigma + (1-\theta)(2D+\sigma)}{\theta}\,\mu_s.
\end{aligned}$$

The ratio bound follows from $\tau_s = \sigma\mu_s$. $\square$

This lemma is the genuine progress in the fixed-anchor idea. It is unconditional and uses only monotonicity, maximality, and the fixed anchor.

4 A conditional $O(\varepsilon^{-1})$ theorem

The previous section reduces the open problem to a specific strong-solver question. We state the required hypothesis explicitly.

Assumption 4.1 (Residual-warm-start strong-solver bound)**.** For the family of anchored strongly monotone inclusions

$$0 \in T_\mu(x) = F(x) + B(x) + \mu(x - a), \qquad 0 < \mu \le \mu_0,$$

there exist constants $A_0, A_1 > 0$, independent of μ, R_{in}, and τ, with the following property. Given a starting point $y \in \operatorname{dom} B$ and a computable certificate

$$r \in T_\mu(y), \qquad \|r\| \le R_{\text{in}},$$

the strongly monotone method returns $x \in \operatorname{dom} B$ and $e \in T_\mu(x)$ with $\|e\| \le \tau$ after at most

$$A_0 + A_1 \frac{1}{\mu} \log^+ \left(\frac{R_{\text{in}}}{\tau} \right)$$

fundamental operations.

Here $\log^+(t) := \max\{0, \log t\}$. The constants may depend on the line-search parameters and on local Lipschitz constants on a bounded set containing the iterates and trial points of the strongly monotone method. The important point is that the logarithm is controlled by the supplied initial *residual certificate*, not by a cold-start distance bound.

Theorem 4.2 (Conditional fixed-anchor continuation bound)**.** *Assume Assumption 4.1. Suppose the first anchored problem with parameter μ_0 can be solved to tolerance $\tau_0 = \sigma\mu_0$ in a number of operations independent of ε. Then the following fixed-anchor continuation scheme returns an ε-residual solution using $O(\varepsilon^{-1})$ fundamental operations.*

At stage s, solve

$$0 \in T_{\mu_s}(x), \qquad \mu_s = \mu_0 \theta^s,$$

to obtain

$$e_s \in T_{\mu_s}(x^s), \qquad \|e_s\| \le \sigma\mu_s.$$

Stop once the explicit original residual certificate

$$q_s := e_s - \mu_s(x^s - a) \in T(x^s)$$

satisfies $\|q_s\| \le \varepsilon$.

Proof. Correctness is immediate: if $\|q_s\| \le \varepsilon$ and $q_s \in T(x^s)$, then $\operatorname{res}_T(x^s) \le \varepsilon$.

By Lemma 2.2, every stage output satisfies

$$\operatorname{res}_T(x^s) \le 2(D + \sigma)\mu_s.$$

Therefore the stopping test must succeed by the first index K_ε for which

$$\mu_{K_\varepsilon} \le \frac{\varepsilon}{2(D + \sigma)}.$$

Equivalently,

$$K_\varepsilon \le \left\lceil \frac{\log(2(D + \sigma)\mu_0/\varepsilon)}{\log(1/\theta)} \right\rceil_+ .$$

For $s \ge 1$, Lemma 3.1 gives an initial certificate for the s-th subproblem with

$$R_{\text{in,s}} \le C_{\text{w}} \mu_s, \qquad \tau_s = \sigma\mu_s.$$

Thus Assumption 4.1 yields a stage cost

$$N_s \le A_0 + A_1 \frac{1}{\mu_s} \log^+ \left(\frac{C_{\mathrm{w}}}{\sigma} \right) \le B_0 + \frac{B_1}{\mu_s}$$

with constants independent of s and ε. Including the first-stage cost in the constant,

$$\begin{aligned} \sum_{s=0}^{K_\varepsilon} N_s &\le C_0 + C_1 \sum_{s=0}^{K_\varepsilon} \frac{1}{\mu_0 \theta^s} \\ &= C_0 + \frac{C_1}{\mu_0} \frac{\theta^{-K_\varepsilon - 1} - 1}{\theta^{-1} - 1} = O(\theta^{-K_\varepsilon}). \end{aligned}$$

Since $\mu_0 \theta^{K_\varepsilon} = \Theta(\varepsilon)$, we have $\theta^{-K_\varepsilon} = O(\varepsilon^{-1})$. Hence the total operation count is $O(\varepsilon^{-1})$. □

Remark 4.3 (Meaning of the conditional theorem). Theorem 4.2 is a reduction, not a solution of Lu–Mei's open problem. It says that fixed-anchor continuation would remove the logarithm if the strongly monotone line-search method admitted a residual-warm-start complexity theorem. The next section explains why the published Lu–Mei theorem does not supply this theorem.

5 Why this does not prove Lu–Mei's open problem

Lu–Mei's strongly monotone theorem is not stated in the residual-warm-start form of Assumption 4.1. In the notation of their Theorem 2, the iteration bound contains a factor of the form

$$\frac{\log \left(C\, r_0 / \delta \right)}{\log(1 + c\mu)},$$

where r_0 is an initial distance to the exact solution of the strongly monotone subproblem, δ is the requested residual tolerance, and $C, c > 0$ are constants determined by the line-search parameters and local Lipschitz constants. Since $\log(1 + c\mu) = \Theta(\mu)$ for small μ, this is a distance-based complexity bound of the schematic form

$$O\left(\frac{1}{\mu} \log \frac{r_0}{\delta} \right).$$

In the fixed-anchor scheme, the target tolerance at stage s is

$$\delta = \tau_s = \sigma \mu_s.$$

The residual-transfer lemma gives a residual certificate of size $O(\mu_s)$, but the distance from the initial point x^{s-1} to the new exact solution p_{μ_s} is only bounded by a constant from Lemma 2.2: indeed

$$\left\| x^{s-1} - p_{\mu_s} \right\| \le \left\| x^{s-1} - a \right\| + \| p_{\mu_s} - a \| \le 4D + \sigma.$$

Using only this information in Lu–Mei's theorem gives

$$N_s \le O\left(\frac{1}{\mu_s} \log \frac{1}{\mu_s} \right).$$

Summing over $\mu_s = \mu_0 \theta^s$ up to $\mu_s = \Theta(\varepsilon)$ gives

$$\sum_{s \le K_\varepsilon} \frac{1}{\mu_s} \log \frac{1}{\mu_s} = O\left(\varepsilon^{-1} \log \varepsilon^{-1} \right).$$

This is exactly the known rate. Therefore the fixed-anchor argument, combined only with Lu–Mei's published Theorem 2, does not solve the problem.

Proposition 5.1 (Residual warm starts are not the same as distance warm starts)**.** *There are locally Lipschitz monotone examples in which consecutive fixed-anchor subproblems have constant-factor residual warm starts but not constant-factor distance warm starts at the target scale $\tau_s = \Theta(\mu_s)$.*

Proof. Take $n = 1$, $B = 0$, $F(x) = x^3$, and anchor $a = 1$. Then F is monotone and locally Lipschitz, and $T = F$ has the solution $x^\star = 0$. The exact anchored solution p_μ is the unique root in $(0, 1)$ of

$$x^3 + \mu(x - 1) = 0,$$

that is,

$$p_\mu^3 = \mu(1 - p_\mu).$$

Writing $p_\mu = \mu^{1/3} y_\mu$, we get

$$y_\mu^3 + \mu^{1/3} y_\mu = 1,$$

so $y_\mu \to 1$ as $\mu \downarrow 0$. Hence

$$p_\mu \sim \mu^{1/3}.$$

Let $\nu = \mu/\theta$ with fixed $0 < \theta < 1$. The old exact solution p_ν has residual for the new μ-subproblem

$$T_\mu(p_\nu) = p_\nu^3 + \mu(p_\nu - 1) = \nu(1 - p_\nu) - \mu(1 - p_\nu) = (\nu - \mu)(1 - p_\nu) = O(\mu).$$

Thus it is a constant-factor residual warm start for tolerance $\Theta(\mu)$. But

$$|p_\nu - p_\mu| \sim \mu^{1/3}(\theta^{-1/3} - 1),$$

so

$$\frac{|p_\nu - p_\mu|}{\mu} \to \infty.$$

The distance warm start is not at the target scale. A theorem whose logarithm uses the distance divided by $\Theta(\mu)$ therefore still sees a growing logarithm on this example. □

The example is not a lower bound against all algorithms. It only demonstrates the logical gap: a residual certificate of size $O(\mu)$ does not automatically imply an initial-distance estimate strong enough to remove the logarithm from a distance-based theorem.

6 Where the proposed proof fails

The proposed proof being checked contains exactly the correct fixed-anchor residual-transfer idea, but it treats Assumption 4.1 as if it were already proved by Lu–Mei's strongly monotone theorem. That is the serious mathematical error. Lu–Mei's theorem supplies a bound in terms of the initial distance r_0 and the requested residual tolerance. It does not prove that the algorithm exploits a supplied initial residual certificate so that the stage cost is

$$O\left(\frac{1}{\mu_s} \log \frac{O(\mu_s)}{\sigma \mu_s}\right) = O\left(\frac{1}{\mu_s}\right).$$

Without that missing residual-warm-start theorem, the proof reverts to

$$O\left(\frac{1}{\mu_s} \log \frac{1}{\mu_s}\right)$$

per stage and hence to the known total complexity

$$O(\varepsilon^{-1}\log\varepsilon^{-1}).$$

There is also a secondary technical point. Any complete proof based on a locally Lipschitz line search must verify a uniform compact set containing not only the outer continuation points but also all inner iterates and line-search trial points. The boundedness lemmas above control the exact path and approximate outer points. They do not, by themselves, prove the required uniform inner-iterate containment for a particular algorithm. Lu–Mei handle this kind of issue in their own analysis by constructing enlarged sets on which local Lipschitz constants are finite. A full proof of Assumption 4.1 would need to do the same.

7 Conclusion

The fixed-anchor Tikhonov continuation method gives a real and useful reduction. The exact and approximate regularized solutions stay uniformly bounded, and the residual-transfer identity

$$e_s^{\mathrm{in}} = e_{s-1} - (\mu_{s-1} - \mu_s)(x^{s-1} - a)$$

shows that a solved μ_{s-1}-subproblem supplies an $O(\mu_s)$-residual warm start for the next μ_s-subproblem. This would remove the logarithm if the strongly monotone locally Lipschitz method had residual-warm-start complexity.

At present, the argument should be classified as *partial progress*, not a solution. The missing mathematical task is to prove Assumption 4.1 for Lu–Mei's strong solver or to design another locally Lipschitz strong solver with that residual-warm-start guarantee and verifiable residual certificates. Until that is done, the open $O(\varepsilon^{-1})$ problem remains unsolved by this approach.

References

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